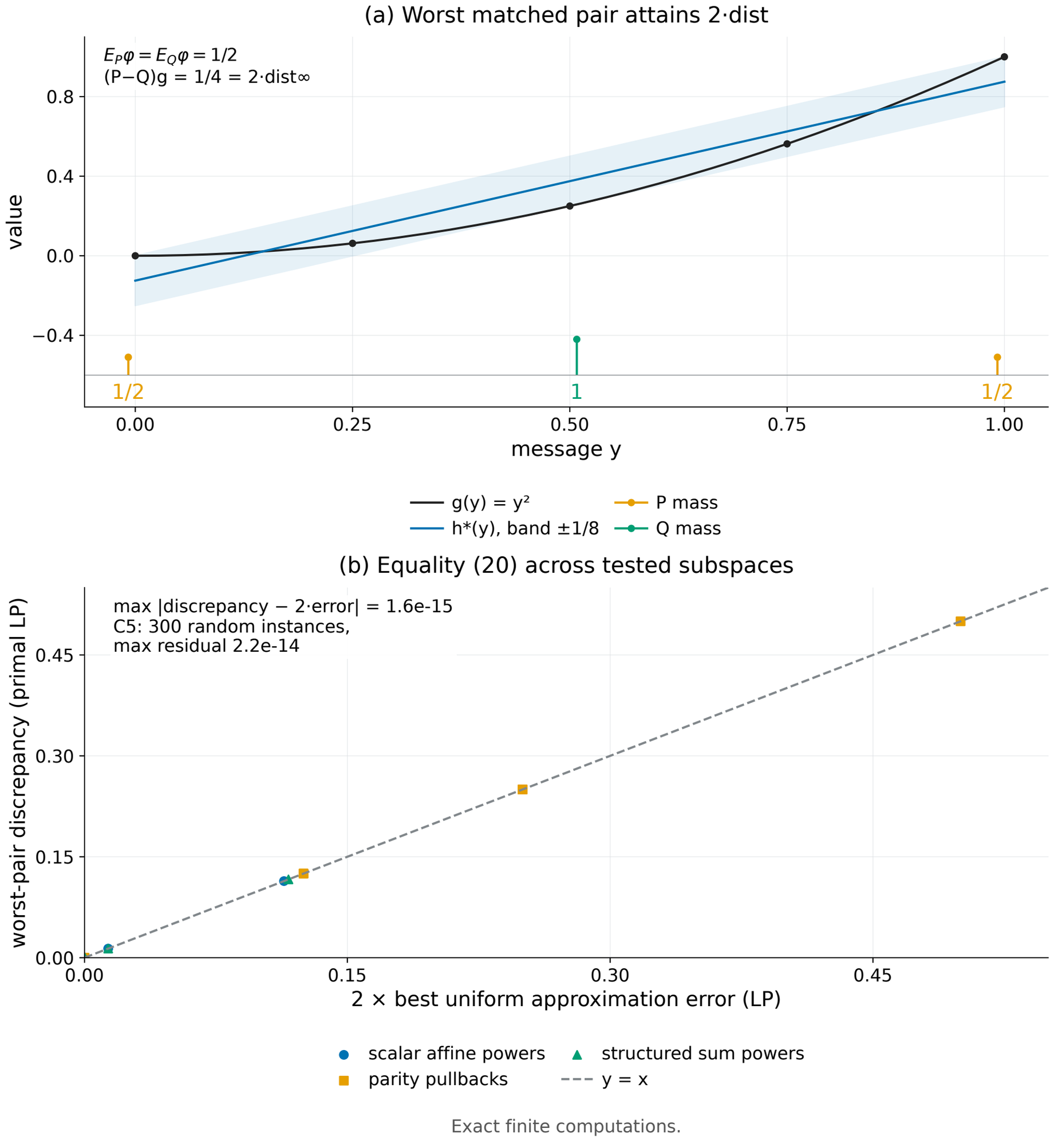
(a) Worst matched pair attains 2·dist
$E_P\varphi = E_Q\varphi = 1/2$
(P−Q)g = 1/4 = 2·dist∞
value
0.8
0.4
0.0
−0.4
1/2
1
1/2
0.00
0.25
0.50
0.75
1.00
message y
g(y) = y²
h*(y), band ±1/8
P mass
Q mass
(b) Equality (20) across tested subspaces
max |discrepancy − 2·error| = 1.6e-15
C5: 300 random instances,
max residual 2.2e-14
worst-pair discrepancy (primal LP)
0.45
0.30
0.15
0.00
0.00
0.15
0.30
0.45
2 × best uniform approximation error (LP)
scalar affine powers
structured sum powers
parity pullbacks
y = x
Exact finite computations.